

Jose Carranza (Mike)

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Education

Purdue University

- Electrical Engineering B.S **3.47 GPA**
- MSECE – Master’s in Electrical and Computer Engineering: ML & AI **3.65 GPA**

West Lafayette, Indiana

August 2022 – December 2025

January 2026 – Present

Lamatepec School

High School Diploma & International Baccalaureate in CS. **3.9 GPA**

La Libertad, El Salvador

June 2021

Programa Jóvenes Talento El Salvador

- *Competitor & student:* **International Mathematical Olympiad** 2019-2020-2021

San Salvador, El Salvador

March 2013 - July 2021

Experience

Boiler Quant Finance Group

Quantitative Analyst

West Lafayette, Indiana, US

August 2025- December 2025

- Built concurrent link-discovery crawlers for a multi-sportsbook odds pipeline (crawl4ai, asyncio), batch-crawling sites and classifying betting event pages via regex URL pattern sets with domain filtering and deduplication.
- Wrote Playwright scrapers using AgentQL semantic queries to extract player-prop lines and over/under prices from dynamically rendered pages, with DOM-interaction fallbacks for expandable markets, exporting structured CSVs for +EV analysis.

Morgan Stanley

Quantitative Finance Strategist Intern

Manhattan, New York, US

June 2nd, 2025-August 8th, 2025

- Evaluated 15 internal fixed-income datasets for external sale in collaboration with dataset-owner strategists. Wrote kdb/q queries to remove third-party-licensed sources and shield proprietary curves, shipping 3 datasets to production and later adding a volume-weighted FX volatility metric.
- Compiled dataset ownership, source, and lineage metadata into a JSON catalog and scripted analysis of a legacy visualization tool's query layer, consolidating it to ~20 master tables to scope a platform migration.
- Built a Python proof-of-concept API providing clients controlled, sampled access to the datasets and received a return offer.

Daimler Truck AG

Powertrain Mechatronics Intern

Portland, Oregon, US

May, 2024-December, 2024

- Built automated test-procedure tools in Simulink for CAN signal processing in powertrain regression testing, adopted by system feature owners as an ECU testbench.
- Designed a master breakout box supporting multiple generations of a powertrain controller, plus individual test harnesses enabling each engineer to bench-test the controller independently.
- Migrated test packages to the ECU-Test automation platform under expanding technical scope.

Purdue University STARS Program

Summer Training Awareness and Readiness for Semiconductors Intern

West Lafayette, Indiana, US

May 15th, 2023-July 7th, 2023

- Designed and implemented digital logic on FPGAs, including adders (RCA, CLA, BCD), an ALU, finite state machines, counters, and shift registers, in a semiconductor-focused summer program.

Stellantis (Fiat Chrysler Automotive)

EE Validation Engineering Intern

Auburn Hills, Michigan, US

August, 2023-Decemeber, 2023

- Executed DVP&R test procedures validating requirements on electrical components and subsystems, proposing improvements to both procedures and requirements.
- Interfaced with ECUs and sensors to collect and analyze test data, presenting findings and failure documentation to Design Release Engineers.

Pacific Coast Companies, Inc.

Data & analytics Intern

Rancho Cordova, California, US

April 18th, 2022-August 5th, 2022

- Built SKU demand forecasting integrated into SAP using ARIMA, SARIMA, and Facebook Prophet in Python and R, adapting Prophet's logistic-growth mode to cap forecasts at 20% above historical peak demand, and classifying demand patterns in SQL by variance and demand spacing.
- Delivered Power BI analyses adopted for inventory planning across subsidiaries.

Academic Experience

Purdue University

West Lafayette, IN, US

ECE 36800: Data Structures Undergraduate Teaching Assistant

September 4th, 2025- December 2025

- Taught algorithm analysis and core data structures (balanced trees, heaps, graph traversal, shortest paths, hashing) to undergraduates, debugging student C assignment implementations in office hours.

ECE 5700: AI - Deep Learning for High-Frequency Crypto Forecasting

August 2025- December 2025

- Reproduced ICLR 2025 results on spectral bias in state space models: implemented a diagonal S4D layer from the paper's equations, with FFT-domain convolution and a Sobolev frequency filter, in PyTorch.
- Extended the paper by making the frequency-tuning parameters learnable and adding a residual forecasting head predicting price changes rather than levels, resolving scale non-stationarity in BTC-USD daily data.
- Ran multi-seed ablations across the tuning grid, identifying a stability boundary: amplifying high frequencies ($\beta > 0$) diverged (MSE > 28) in low-SNR financial data, while dampening ($\beta = -1$) converged, confirming spectral filters must regularize, not enhance, for financial series.

ECE 49022: Senior Design – Patek Digitale Wearable System

August 2025- December 2025

- Designed the power subsystem for a battery-operated wrist wearable: dual-rail architecture with a buck-boost converter (3.3 V) and low-noise LDO (1.8 V) from a single-cell 350 mAh Li-Po, plus a USB-C linear charging circuit at a 0.5C rate with TVS input protection.
- Performed KiCad layout for the power section and the board-level integration routing connecting the power and sensor sections of the mixed-signal PCB.
- Validated the design on lab instruments: measured rail ripple of 44 mV and 24 mV pk-pk against a 50 mV spec, verified 170 mA charge current via the charger's program pin, and confirmed 17+ hours of runtime against a 16-hour requirement from a 77 mW power-budget analysis.

ECE 33700: ASIC Design Laboratory – USB 2.0 to AHB-Lite Subordinate Interface

January 2025- May 2025

- Designed the AHB-Lite subordinate interface for a USB 2.0 full-speed device: implemented a memory-mapped register file and dual FSMs unpacking byte, halfword, and word transfers into sequential accesses to a shared 64-byte packet FIFO.
- Implemented AHB-Lite protocol compliance, including pipelined address and data phases, wait-state insertion via HREADY, and two-cycle HRESP error responses for illegal transfer sizes and addresses.
- Wrote the top-level and module-to-module integration test benches, running directed test cases against the AHB interface and debugging failures at the waveform level and verified modules individually and pairwise on the synthesized gate-level netlist.

Skills & Interests

Technical: Python, C, C++, SystemVerilog, kdb/q, SQL, R, Java, CAPL/CANoe, digital design and verification, AHB-Lite, I2C, FPGA prototyping, KiCad, lab instrumentation, PyTorch, pandas, NumPy, ARIMA/SARIMA, Prophet, Power BI, SAP HANA, Git, Jira.

Language: English, Native Spanish (Advanced Speaking & Writing), Basic German (Reading Proficiency)

Interests: Programming, AI, quantitative finance, Investing, Hardware, Firmware, Olympic lifting, Track & Field.